

AURA / EVT-A03 NATIVE ELECTRICAL DESIGN

Radio and debug

- nRF52840 BLE radio
- SWD programming and recovery pads
- 3.0 V normal-voltage module supply

File: a03-radio.kicad_sch

Audio and privacy

- Dual PDM microphones and bus isolation
- Mechanical microphone power cut
- Hardwired amber microphone-power indicator

File: a03-audio.kicad_sch

Offline storage

- 1 Gbit SPI NAND for offline audio
- Power-up write-protect and hold pull-ups
- Firmware must manage ECC and bad blocks

File: a03-storage.kicad_sch

Charging and thermal limit

- 53 mA nominal charger and system power path
- Pack NTC comparator; default-off charge permission
- Protected 4.2 V cell and qualified NTC required

File: a03-charging.kicad_sch

System power and controls

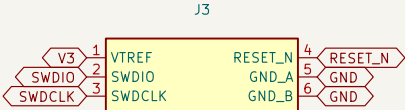
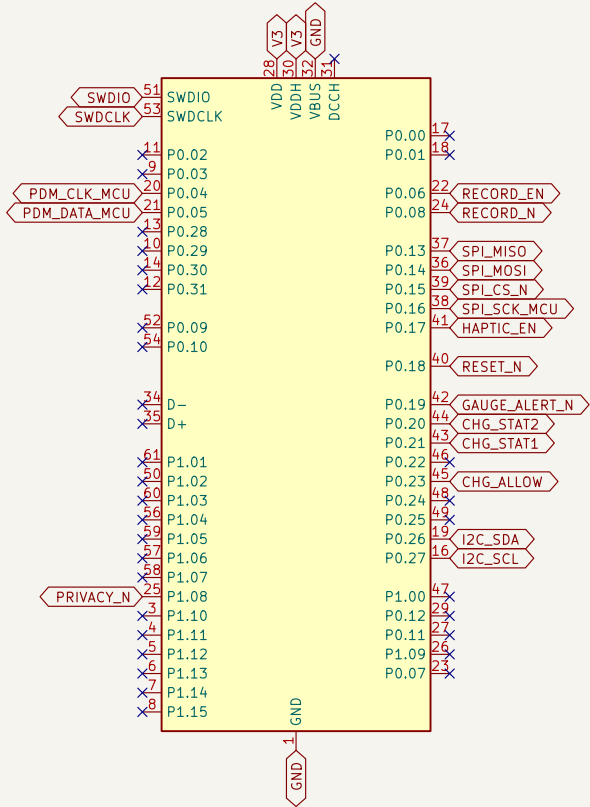
- 3.0 V system rail and battery fuel gauge
- Tactile capture button
- Closed-loop LRA haptic driver

File: a03-controls.kicad_sch

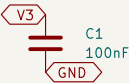
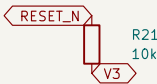
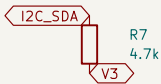
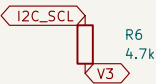
AURA product engineering		
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Title: AURA / Native electrical design		
Size: A4	Date: 2026-09-08	Rev: EVT-A03
KiCad E.D.A. 10.0.4		Id: 1/6

AURA A03 / RADIO + PROGRAMMING

U1
MDBT50Q-1MV2



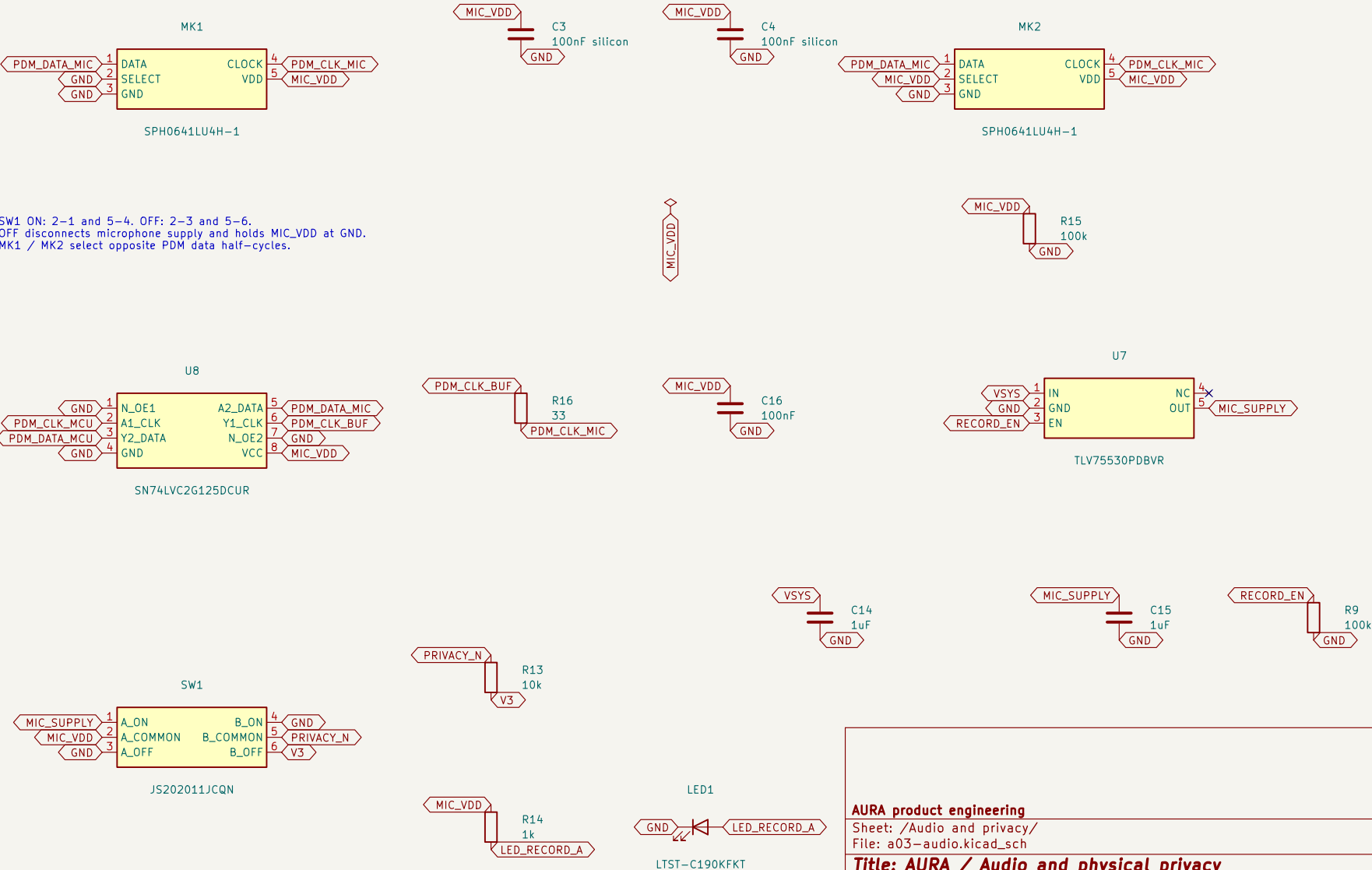
AURA-SWD-PADS



AURA product engineering
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File: a03-radio.kicad_sch
Title: AURA / Radio and programming

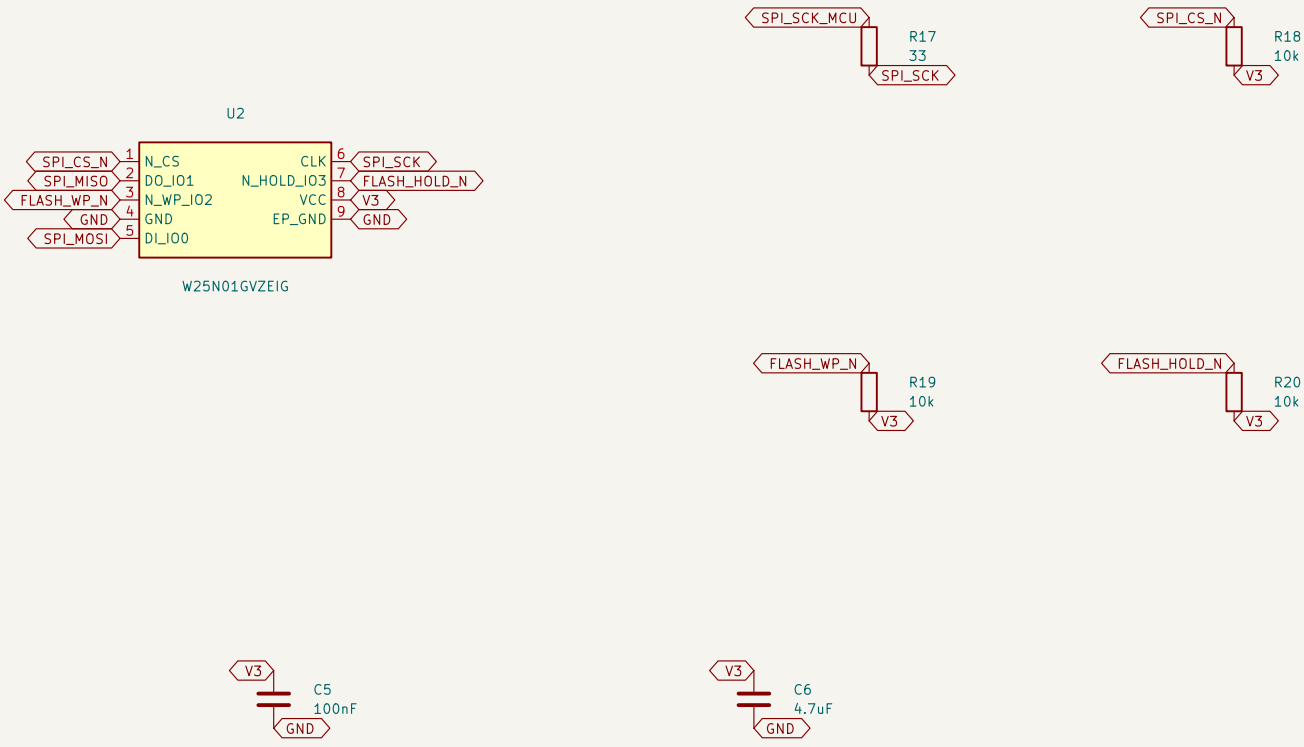
Size: A4	Date: 2026-09-08	Rev: EVT-A03
KiCad E.D.A. 10.0.4		Id: 2/6

AURA A03 / AUDIO + HARDWARE PRIVACY



AURA product engineering	
Sheet: /Audio and privacy/ File: a03-audio.kicad_sch	
Title: AURA / Audio and hardware privacy	
Size: A4	Date: 2026-09-08
KiCad E.D.A. 10.0.4	Rev: EVT-A03 Id: 3/6

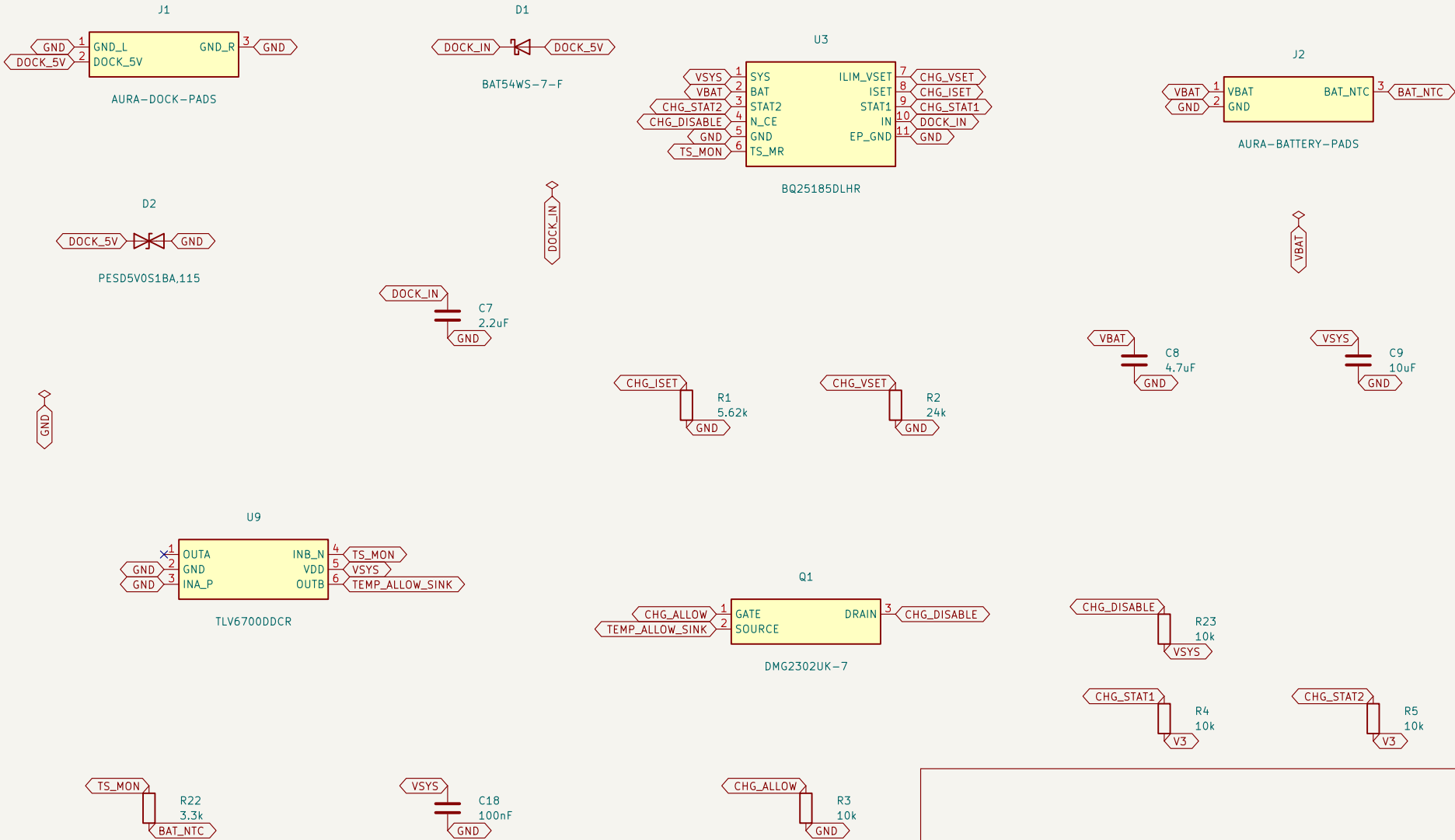
AURA A03 / OFFLINE FLASH STORAGE



ZE package exposed pad pin 9 connects to GND.
Provide NAND ECC, bad-block mapping and wear management.
Audio encryption and crash-safe file recovery belong in firmware.

AURA product engineering		
Sheet: /Offline storage/		
File: a03-storage.kicad_sch		
Title: AURA / Offline flash storage		
Size: A4	Date: 2026-09-08	Rev: EVT-A03
KiCad E.D.A. 10.0.4		Id: 4/6

AURA A03 / CHARGING + THERMAL PERMISSION



U9 OUTB sinks when TS_MON > 0.4 V. Validate with the selected cell and NTC.
Q1 body diode: source pin 2 to drain pin 3. CHG_ALLOW defaults low.
53 mA nominal charge current; bench thermal and pack-fault tests required.

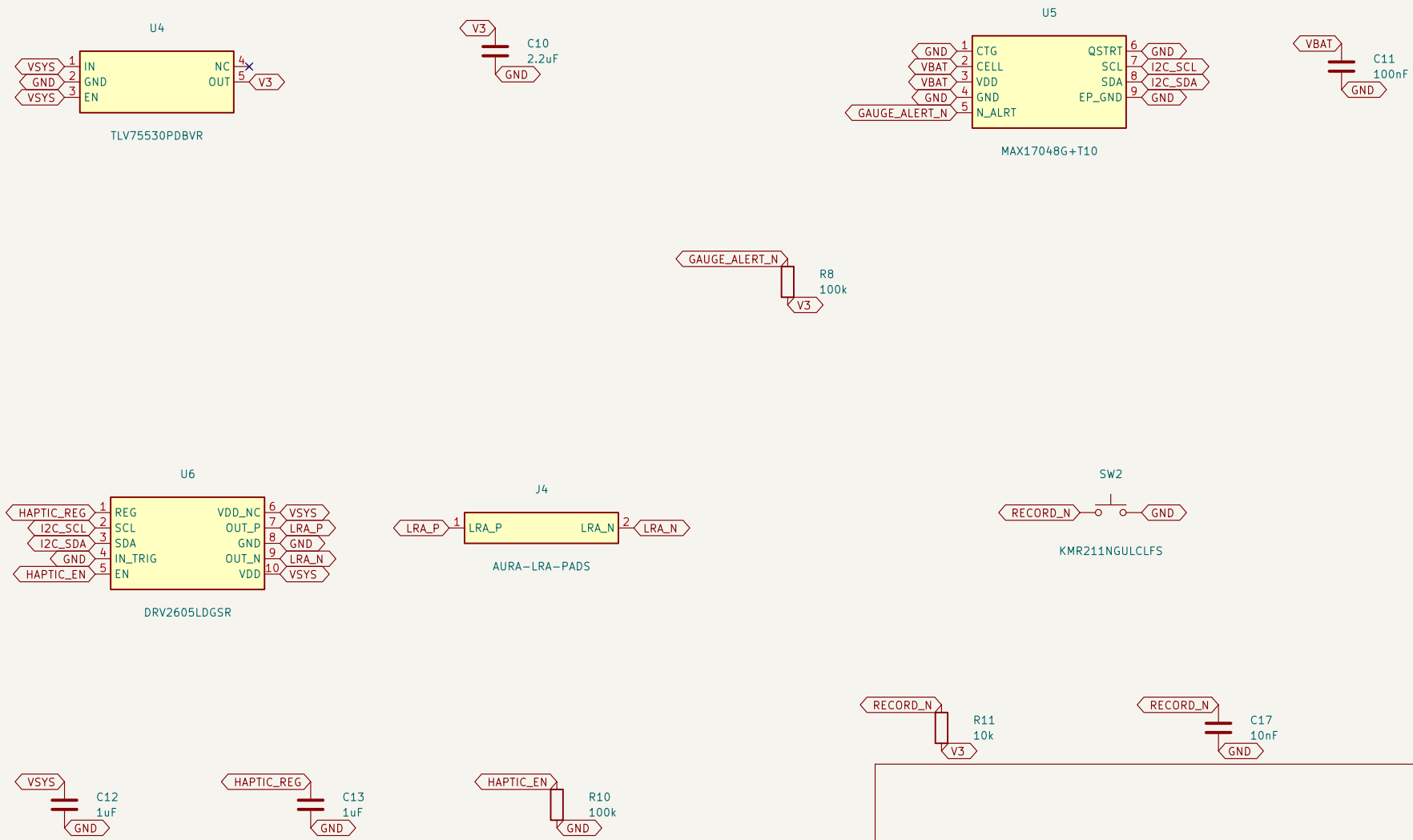
AURA product engineering

Sheet: /Charging and thermal limit/
File: a03-charging.kicad_sch

Title: AURA / Charging and thermal permission

Size: A4	Date: 2026-09-08	Rev: EVT-A03
KiCad E.D.A. 10.0.4		Id: 5/6

AURA A03 / POWER + RECORD + HAPTICS



AURA product engineering		
Sheet: /System power and controls/		
File: a03-controls.kicad_sch		
Title: AURA / System power and controls		
Size: A4	Date: 2026-09-08	Rev: EVT-A03
KiCad E.D.A. 10.0.4	Id: 6/6	